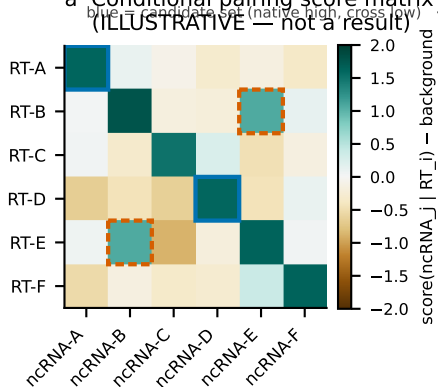
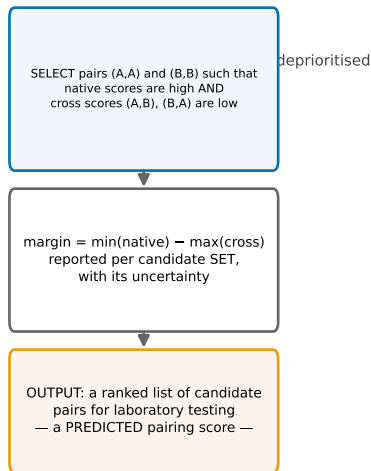


a Conditional pairing score matrix
(ILLUSTRATIVE — not a result)



b Selection rule



c Confidence must be conditioned on

- model uncertainty (seeds, folds, replicates)
- sequence relatedness of the two RTs
- retron type (same type = harder, more informative)
- phylogenetic / lineage distance
- inside the model's training distribution?
- consistency across seeds and cross-fit folds
- native-minus-counterfactual score margin

Most-distant RTs are NOT automatically best: distance itself produces out-of-distribution scores.